Quantitative Finance > Mathematical Finance

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Trading multiple mean reversion

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How should one construct a portfolio from multiple mean-reverting assets? Should one add an asset to portfolio even if the asset has zero mean reversion? We consider a position management problem for an agent trading multiple mean-reverting assets. We solve an optimal control problem for an agent with power utility, and present a semi-explicit solution. The nearly explicit nature of the solution allows us to study the effects of parameter mis-specification, and derive a number of properties of the optimal solution.

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